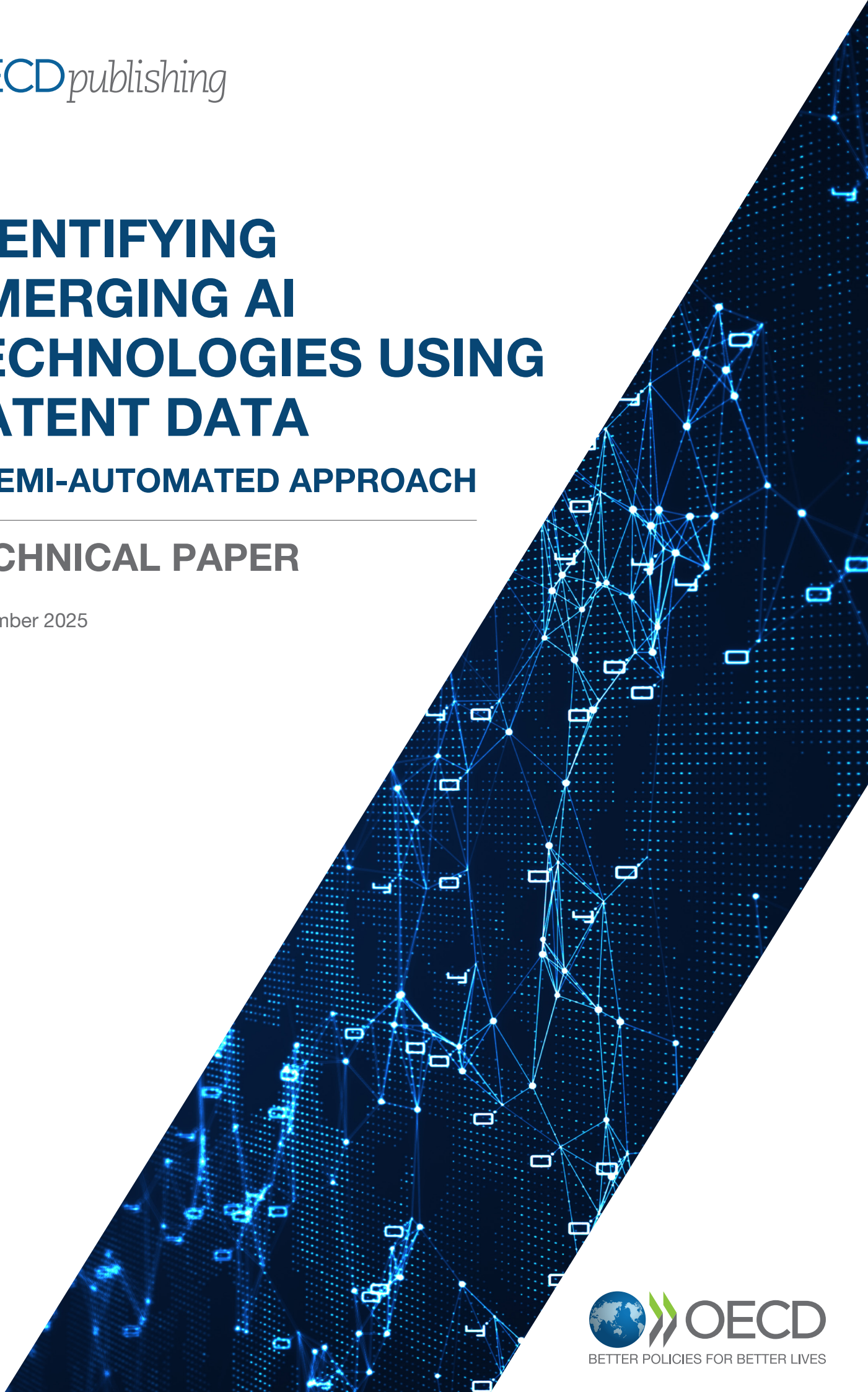
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IDENTIFYING EMERGING AI TECHNOLOGIES USING PATENT DATA

A SEMI-AUTOMATED APPROACH

TECHNICAL PAPER

September 2025



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Abstract

This technical paper introduces a novel methodology for identifying “AI patents”—patents protecting inventions related to artificial intelligence (AI). It builds on existing taxonomies and considers recent developments in the field, notably the surge of generative AI technologies. It combines technology groups of the Cooperative Patent Classification (CPC) with a list of AI-related keywords and phrases. The method for identifying AI vocabulary and CPC classes is designed to enable easy updates and integration of emerging AI technologies.

Keywords: Artificial intelligence; Innovation; Patents

JEL Code: O33, O34

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Executive summary

This document presents an innovative methodology for identifying patents related to artificial intelligence (AI). This methodology enhances the coverage of AI inventions by incorporating the latest terminology and facilitating the inclusion of recent innovations. It is designed for easy updates and adaptability across various patent datasets.

The methodology employs a robust search strategy that leverages the patent classification system, categorising inventions into specific technology fields, and uses a comprehensive list of AI-related keywords extracted from patent titles and abstracts. The identification process to develop the AI patent taxonomy is semi-automated and includes two key steps:

- **Identification of keywords and phrases:** By analysing abstracts of recent scientific articles in AI through text mining on the open access arXiv platform, the process identifies emerging keywords and key phrases of up to three words (e.g., “agentic artificial intelligence”) that reflect the latest advancements in the field.
- **CPC group determination:** The process identifies relevant Cooperative Patent Classification (CPC) groups based on their co-occurrence with AI terms in patent titles or abstracts.

This semi-automated approach allows for regular updates, ensuring the methodology remains aligned with advances in AI and incorporates the most current AI vocabulary and relevant CPC classes. The methodology has been validated by experts through a dedicated survey.

Patents are classified as AI if:

- They fall within one of the five CPC groups identified as *core AI* (G06N3, G06N5, G06N7, G06N20, or G06F18); or
- Belong to a CPC group identified as *AI-related* and contain at least one AI keyword in their title or abstract.

The revised methodology significantly increases the identification of AI patents compared to the previous OECD methodology, particularly by including additional keywords associated with generative AI and image processing techniques. Recent trends in AI patents are analysed using data from Patent Cooperation Treaty (PCT) applications, providing internationally comparable indicators. To address administrative delays in patent publication, recent data is “nowcasted.”

Since 2015, the number of patented inventions in AI technologies has surged, with an average annual growth rate of 30% until 2020. By 2023, AI patents accounted for approximately 6% of all PCT filings – three times the level seen in 2015. From 2020 to 2023, the leading countries AI patent development were the People’s Republic of China (hereafter, ‘China’) (31%), the United States (27%), and Japan (10.5%). However, recent trends indicate a slowdown in AI patenting.

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1 Introduction

This document presents an innovative methodology for identifying “AI patents”, i.e., patents protecting inventions relating to Artificial Intelligence (AI). This revised methodology builds on existing ones, including the previous OECD methodology (Baruffaldi et al., 2020[1]). It considers recent developments in AI technology, notably integrating the surge in generative AI. The proposed methodological approach relies on a search strategy based on patent technology classes, using the Collaborative Patent Classification (CPC) scheme, and AI-related vocabulary.

The revised taxonomy was developed using a semi-automated approach. It identifies emerging AI terminology by text mining recent scientific articles. Lists of CPC groups potentially related to AI are then identified based on their co-existence alongside AI terms in patent titles or abstracts along with expert validation. This approach enables frequent updates of the methodology to incorporate emerging AI technologies promptly. While the analysis relies on patent applications filed under the Patent Cooperation Treaty (PCT) providing internationally comparable and timely indicators, the methodology and resulting taxonomy can be easily applied on other patent datasets.

This report consists of 3 sections. The first section summarises recent AI patent landscaping efforts. The second section describes the revised methodology for identifying AI patents. Finally, the third section highlights recent trends in AI patenting, with a focus on PCT filings.

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2 Review of existing taxonomies

In light of the growing policy, academic and business interest in AI, and of the unique richness of patent data to measure technological inventions, efforts have been undertaken by academia and IP offices to monitor recent developments in the field.

However, it is important to recognise that patent data gives only a partial picture of technological advancements. Not all inventions are protected by patents, and not all inventions are patentable (Griliches, 1990[2]). This limitation is particularly relevant for AI technologies, as patentability can vary by jurisdiction. For instance, while the United States patent office permits the patenting of software, the European Patent Office (EPO) only considers "computer implemented inventions" as patentable (Baruffaldi et al., 2020[1]). Thus, the landscape of AI innovation is shaped not only by patent data but also by the legal frameworks governing patentability.

The partial survey by Hötte et al. (2022[3]) indicates that the outcomes of patent landscaping exercises can vary significantly, largely depending on the definition of AI and the types of patents included. Below is a brief summary of recent efforts to delineate AI patents.

WIPO (2019)

The WIPO query strategy primarily uses classification codes to overcome the limitations of keyword-based searches in broad technological fields like AI (WIPO, 2019[4]). Different classification systems that exist - e.g., the International Patent Classification (IPC), the Cooperative Patent Classification (CPC) or the Japan Patent Office (JPO) file index (FI) or file forming term (F-term) - vary in scope, precision, and structure, requiring tailored approaches. AI patents often lack specific AI classification codes, necessitating the inclusion of broader categories controlled with extended keyword lists to reduce noise. Some patents can only be identified through keywords, especially where classification systems (e.g., FI/F-terms) are restricted to specific jurisdictions like JPO.

The query structure includes:

- **Block 1:** Selected CPC codes specific to AI technologies.
- **Block 2:** AI-specific keywords.
- **Block 3:** Broader CPC/IPC/FI/F-terms classes paired with AI-related keywords.

The final query combines these blocks. Relevance and specificity were tested via Orbit database[1] and manual sampling. Keywords were derived from literature and web resources, with detailed search strings, codes, and keywords provided in the document.

OECD (2020)

In 2020, the OECD developed a methodology to identify patents related to AI using a combined IPC/CPC and keyword-based approach (Baruffaldi et al., 2020[1]). A set of AI-related keywords is first identified via text mining on recent AI-related scientific publications, using Elsevier's SCOPUS© database. A keyword

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search is then applied on patent's titles, abstracts, and claims to delineate IPC or CPC codes that are relevant to AI (either considered as core AI or related to AI). Both AI-related keywords and technology classes have been validated by experts in the field (researchers and/or patent examiners). The analysis was performed on the European Patent Office (EPO) Worldwide Patent Statistical Database (EPO, PATSTAT Global)[2] and on the United States Patent and Trademark Office (USPTO) PatentsView database[3]. The resulting search strategy was performed in three steps:

- Select patents with core AI IPC or CPC classes; and
- Select patents with AI-related IPC or CPC classes that also feature at least one AI-related keyword in its abstract, title and or claim (for USPTO patents only); and
- Select patents that feature *at least* three AI-related keywords.

This search strategy is however dependent on the state of knowledge at a given point in time and does not encompass the more recent developments in the AI field, such as generative AI.

Country Activity Tacker (2020)

The CSET, the Center for Security and Emerging Technology of Georgetown University, publishes the CAT (Country Activity Tacker), which is a database and indicators of AI related patents (Emerging Technology Observatory, 2022[5]). The CAT covers 52 patent jurisdictions and includes data on patent applications and grants.

For identifying AI patents, the CAT mixes technological classifications (CPC and IPC) and keywords (Murdick and Thomas, 2020[6]). It selects about 90 CPC categories at the most granular level and 12 key terms. The search strategy is either based on patent classifications or key terms: a patent that does not refer to an AI CPC category but mentions a term like "artificial intelligence" or "fuzzy logic" is considered as AI. In addition, the CAT proposes data and indicators[4] on 7 sub-fields of AI (Machine Learning, etc.), using various patent data sources.

USPTO (2021 and 2024)

The USPTO relies on machine learning techniques to identify AI patents, as described in Giczy, Pairolero and Toole (2021[7]) and, more recently in Pairolero et al. (2024[8]). The AI patent landscaping method includes the following steps:

- Definition of eight sub-fields of AI: "Machine learning", "Evolutionary computation", "Natural language processing (NLP)", "Vision", "Speech", "Knowledge processing", "Planning and control", and "AI hardware".
- For each subfield, identification of a "seed set" (a set of patents that are assessed as belonging to the field, in view of their technological category), an "antiseed set" (patents that are assessed as not belonging to the field) and an "expansion set" (patents that are candidates for belonging to the field).
- Vectorisation of the textual content of these patents using a transformer (in 2021, Word2Vec technique was applied).
- Training of a classifier for each of the eight categories on the data sets.
- Manual validation of classifier's predictions on a test set.

The resulting AI Patent dataset (AIPD) was constructed on USPTO patents and made publicly available to researchers.

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WIPO on Generative AI (2024)

Generative AI (GenAI) currently lacks established patent classifications. The most relevant CPC category i.e., G06N3/045 (auto-encoder networks), was introduced in 2023, and reclassification of earlier patents into this category is underway.

A tailored two-stage approach was developed by WIPO to capture GenAI patents (WIPO, 2024[9]):

- **Stage 1:** Classical searches combined with AI prompts retrieved an initial high-recall dataset.
- **Stage 2:** A Bidirectional Encoder Representations for Transformers (BERT) classifier refined this dataset, improving precision by distinguishing GenAI patents from others that use AI techniques.

Additionally, three methods were used to build the dataset:

- **Method 1a:** Generic searches for "generative AI" concepts across classifications.
- **Method 1b:** Specific searches for key technologies, e.g., Generative adversarial network (GAN), Large language models (LLMs), Diffusion Models.
- **Method 2:** Around 100 AI-generated prompts targeting GenAI concepts.

The datasets were fine-tuned using a BERT classifier to categorise patents[5] into GenAI patents and non-GenAI patents. The resulting set of GenAI patents is available for download.

Other existing taxonomies

Additional patent analyses have been undertaken by IP offices in the recent years. The Intellectual Property Office of the United Kingdom (UKIPO) released a report on patenting in AI (UKIPO, 2019[10]), building on the work undertaken by WIPO (2019[4]) and OECD (Baruffaldi et al., 2020[1]), with further insights from Canada Intellectual Property Office (CIPO).

In 2020, technologies in the realm of the 4th industrial revolution (4IR), as identified by the EPO, included an AI component, that was also delineated with a combined technology class and keyword search approach (EPO, 2020[11]).

More recently, the IP office of Australia released an interactive patent analytics visualisation focusing on AI patents (IP Australia, 2024[12]), using a mixed IPC and keyword search approach. Their patent landscaping drills down into specific technological domains (e.g., Machine learning, Expert systems, Hardware), and application domains (e.g., Chemistry, Mechanical engineering, Electrical engineering, Instruments, …).

Also released in 2024, a JPO report presents recent trends in AI-related inventions, distinguishing the evolution of patents protecting core AI technologies vs. applications of AI. The definition relies on JPO's file index (FI) system, in particular the class G06N, and a set of keywords (JPO, 2024[13]).

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3 A novel approach to identifying emerging AI fields

The methodology for identifying AI patents proposed in this document builds on existing frameworks, notably the previous OECD methodology described in Baruffaldi et al. (2020[1]). It relies on two key elements: a pre-defined set of Cooperative Patent Classification (CPC) codes and an evolving list of AI keywords. This approach systematically integrates recent technological developments, enriching the vocabulary to capture emerging AI fields. To stay current, it uses a screening process that identifies emerging AI-related terms from recent scientific papers. This process allows for frequent updates of the methodology to ensure it stays relevant. Additionally, this process overcomes the limitations of the previous OECD taxonomy, which excluded recent technologies like generative AI that were not prominent at the time.

The revised methodology combines CPC categories at an aggregate level, as higher CPC levels are less likely to be affected by classification changes, along with AI-related keywords. This paper builds on patent applications filed under the Patent Cooperation Treaty (PCT) to provide timely and internationally comparable indicators.

Enrich the list of AI keywords to capture emerging trends

Keywords can help identify emerging AI developments and techniques. They are found in the title or the abstract of patent documents, which are typically 150 to 200 words long. To keep up with the rapid evolution of AI, it is crucial to maintain up-to-date AI keyword lists for patent analysis. The revised AI keyword list proposed in this paper is built by combining two elements:

- A “base” keyword list derived from the one agreed upon by experts commissioned by the OECD (Baruffaldi et al., 2020[1]) and from the list derived from WIPO’s report on AI patents (WIPO, 2019[4]); and
- Complementary AI keywords extracted from the abstracts of AI papers published on arXiv[6] since 2020, to identify emerging terminology. This process uses text mining techniques described in Box 3.1.

arXiv is a free, open-access archive that collects a wide range of scholarly content, including articles, academic theses, conference proceedings, books or book chapters, lecture notes, and translations. Content is categorised by field of study or scientific disciplines according to the arXiv taxonomy. While arXiv allows authors to self-classify their submissions with feedback from moderators, the material is not necessarily peer-reviewed, unlike other established databases such as Elsevier’s SCOPUS. However, it is assumed that the terminology used by authors reflects the latest scientific developments. Furthermore, as arXiv is fed directly by researchers, with no delay in availability, it is reflective of the latest trends in the domain.

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Box 3.1. Systematic identification and extraction of recent AI keywords

AI keywords are extracted from the arXiv metadata dataset, which comprises entries for each scientific paper submitted. These entries include bibliographic information such as the paper identifier, submitter, authors, title, comments, journal reference, doi, abstract, categories, and versions. The dataset, available on Kaggle,7 is curated and updated by arXiv on a weekly basis.

The following steps were implemented to extract AI keywords from arXiv:

- **Selecting abstracts:** Abstracts from 2020 onwards were collected for all submissions classified under "AI" as the main category. arXiv organises AI subfields into different subject areas: the category “cs.AI” covers most AI-related areas, except *Vision*, *Robotics*, *Machine Learning*, *Multiagent Systems*, and *Computation and Language* (*Natural Language Processing*). About 130,000 abstracts from both “cs.AI” and any of these subfields were extracted.
- **Preparing text data:** The text was processed through tokenisation, stopword removal, alphanumeric character retention, and stemming. Tokenisation refers to the segmentation of text into smaller units called tokens, typically based on whitespace. Stopwords enclose common words like "the" and "a" that contain meaningless information. Stemming involves reducing words to their base or root form, such as "balanced" becoming "balanc," which aids in consolidating variations of words with similar meanings.
- **Extracting recent key AI phrases:** For each publication year, the top 20 most frequent bigrams (sequence of two words, e.g., “machine learning”) and the top 20 most frequent trigrams (e.g., “artificial neural networks”) were identified. Most frequent unigrams (single words) are excluded as they are context-sensitive (e.g., the word "artificial" is not exclusively related to "artificial intelligence"), with exceptions such as “ai”, “llm” or “embedding”.
- **Final list of recent AI keywords:** For the purposes of this study, emerging AI keywords include key AI phrases made up of one, two, or three words (i.e., unigrams, bigrams or trigrams, respectively).

This extraction process is to be conducted twice a year to align with EPO PATSTAT database updates and provide an up-to-date view of AI developments.

Use of patent classification system

A set of technological patent classes was identified according to their relevance to AI. This revised methodology for identifying AI patents uses the CPC system, which is more granular than the International Patent Classification (IPC) system, improving patent searches.[8] The CPC is an international classification system for patent documents which captures their technical content or application. It is hierarchical and contains more than 270 000 categories at the most granular level. CPC categories are assigned to each patent application by the recipient patent office; each patent can have several CPC categories, reflecting the interdisciplinary nature of inventions (many patents have more than 10 categories).

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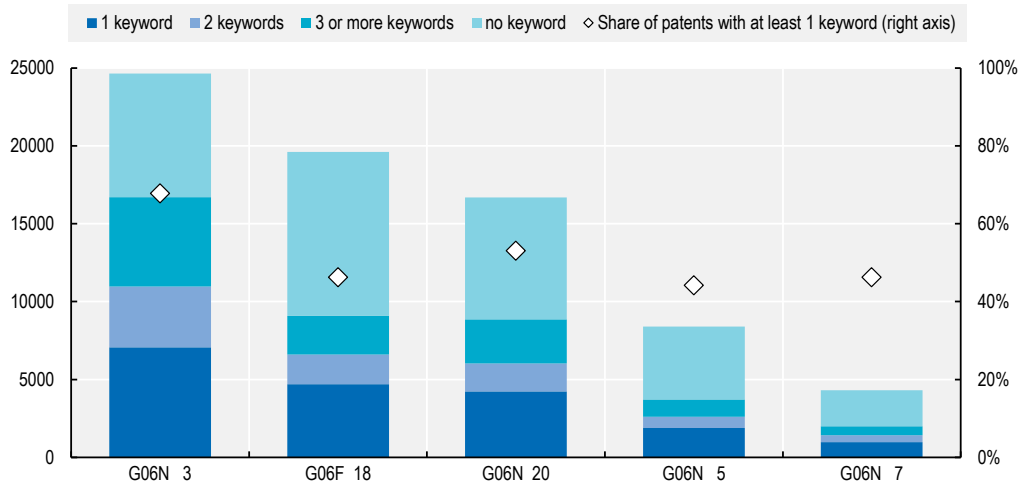
CPC groups related to AI were identified through the following steps:

i. An initial list of AI-related CPC categories was created based on subclass definitions (see Table A A.1), covering the following subclasses:
   - **G06** "*Instruments: Computing; calculating or counting*": G06F, G06J, G06K, G06N, G06Q, G06T, G06V.
   - **G16** "*Instruments: Information and communication technology [ICT] specially adapted for specific application fields*": G16B, G16C, G16H, G16Y, G16Z.
   - **B60** "*Transporting: vehicles in general*": B60W.
   - **B64** "*Transporting: aircraft; aviation; cosmonautics*": B64U.

   Patent applications in these subclasses were further analysed by searching for AI-related terms in their title or abstract.[9]

ii. Selecting "core" AI categories:
   - For each group within these subclasses, the percentage of patents filed between 2018 and 2021 containing 1, 2, or 3 AI keywords in their abstract or title was calculated (see Table A A.2). The AI keyword search was conservative, requiring exact matches and not permitting any additional words inserted between the constituent tokens of keywords (except for the key phrases associated with the term "Transformer" – such as "attention transformer", "language transformer" or "learning transformer" that required more flexibility as the two terms are not always written contiguously[10]). This is done to minimise the risk of including irrelevant patents in the analysis ("false positives").
   - Certain groups (G06N3, G06N5, G06N7, G06N20 and G06F18) showed a strong AI focus: more than 55% of patents in these groups contained at least one AI keyword in their title or abstract; 29% contained at least 2 keywords; and 17% contained at least 3 keywords (Figure 3.1). Since these percentages were significantly higher than in other CPC groups, all patents in these five groups were classified as AI patents.[11]

Figure 3.1. Distribution of patents by "Core AI" CPC groups and AI-related keywords



Note: Data refer to patent applications filed under the Patent Cooperation Treaty (PCT) between 2018 and 2021, in selected CPC groups, using simple counts.
Source: OECD, STI Micro-data Lab: Intellectual Property Database, http://oe.cd/ipstats, January 2025.

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It is important to note that AI keywords may be absent from a patent's title or abstract but still present in its list of claims – which define the subject matter protected by the patent - or in the full text of the patent document. In addition, as patents are legal instruments, the patent language reflects not only technical content but also of legal and strategic considerations. Human experts (patent examiners) might recognise as AI patents certain documents that do not feature AI keywords in their title or abstract: Hence our choice to fully tag such patents as AI according to the above-mentioned AI-related CPC groups.

iii. Additional CPC groups with at least 20 patents filed between 2018 and 2021, where at least 10% contain at least one AI-related keyword in the abstract or title, are also included (see the list of additional CPC groups in Table A A.2).

Noteworthy, using broader CPC groups, rather than more specific categories, makes it easier to update AI patent indicators. This approach minimises disruptions caused by classification changes, which typically occur at more granular levels and would require frequent adjustments to the methodology.

Expert validation

A consultation with experts was conducted to validate and refine the initial selection of CPC groups and keywords. Experts were identified through the OECD Network of Experts on AI (ONE AI) and the OECD-led IP Statistics Taskforce, including patent examiners specialised in AI. The validation exercise involved online surveys to assess the relevance of selected CPC groups and AI-related vocabulary. Experts also reviewed a sample of patents to assess their connection to AI technologies (see Annex C for further details on the consultation process).

Survey results helped refine the list of AI-related keywords, removing those that were deemed insufficiently relevant to AI. CPC groups were validated through a dedicated survey and a review of a random sample of patent abstracts.

The revised methodology for identifying AI patents therefore includes:

- 5 "core AI" CPC groups.
- 95 "AI-related" CPC groups, where patents are considered AI only if they contain at least one AI keyword in the title or abstract.
- A revised AI vocabulary of 274 keywords, to be updated biyearly.

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Revised AI patent identification methodology

The revised methodology for identifying AI patents includes patents that:

- **Belong to "core AI" groups:**
  - **G06N3** (Computing arrangements based on biological models),
  - **G06N5** (Computing arrangements using knowledge-based models),
  - **G06N7** (Computing arrangements based on specific mathematical models),
  - **G06N20** (Machine learning),
  - **G06F18** (Electric digital data processing: pattern recognition);
- Or belong to an **"AI-related" CPC group** (Table A B.2) and include **at least one AI keyword** (Table A B.1) in the title or abstract. This group includes technologies that apply AI to specific uses (e.g., autonomous driving). Combining keywords with CPC groups helps avoid overestimating AI patents by filtering out those not inherently AI-related.

Patents that do not meet either condition are not classified as AI patents.

A comparison was conducted between patent statistics from the 2020 OECD taxonomy (Baruffaldi et al., 2020[1]) and indicators from the revised AI patent identification methodology (see Annex D): the number of AI patents increased by over 50% under the revised methodology on average for the period from 2000 onwards, with nearly 60% of patents identified by both methodologies. The differences mainly stem from the inclusion of additional AI keywords, in particular those related to generative AI techniques and image processing.

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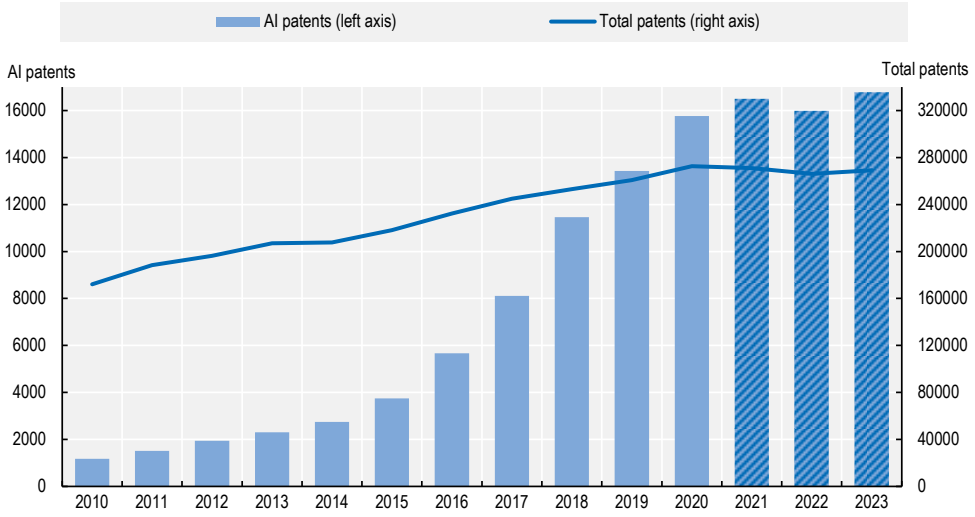
4 Recent trends in AI patents

This section proposes an overview of recent trends in terms of AI technological developments, as observed using patent data extracted with the revised methodology. To ensure relevance and cross-country comparability, the indicators presented below are calculated using patent applications filed under the Patent Cooperation Treaty (PCT), according to the priority year as reference (the closest date to the invention date), and the inventors as a geographical reference (in case of inventors from multiple countries, fractional counts are applied). The analysis follows methodological recommendations from the OECD Patents Statistics Manual (OECD, 2009[14]). Raw data from 2021 onwards are incomplete, as indicators are calculated according to the earliest date at which the invention has been protected by a patent and given the administrative delays observed in IP offices before patents are published. Hence, indicators for the period 2021-23 are estimated using complementary data, following the "nowcasting" procedure described in Annex E.

Evolution of patenting in AI

The number of patented inventions in AI technologies increased rapidly during the last decade, as shown in Figure 4.1, at an average annual growth rate of 30% between 2015 and 2020. In 2023, almost 17 000 AI inventions were filed via the PCT, a volume eleven times larger than that observed in 2011. As a comparison, the number of ICT related patents grew at yearly average of 3% since 2015, following global patents filings trend. Additionally, more than 6% of PCT patents filed embed an AI component in 2023, over 3 times more than in 2015.

Figure 4.1. Evolution of AI patents, 2010-2023



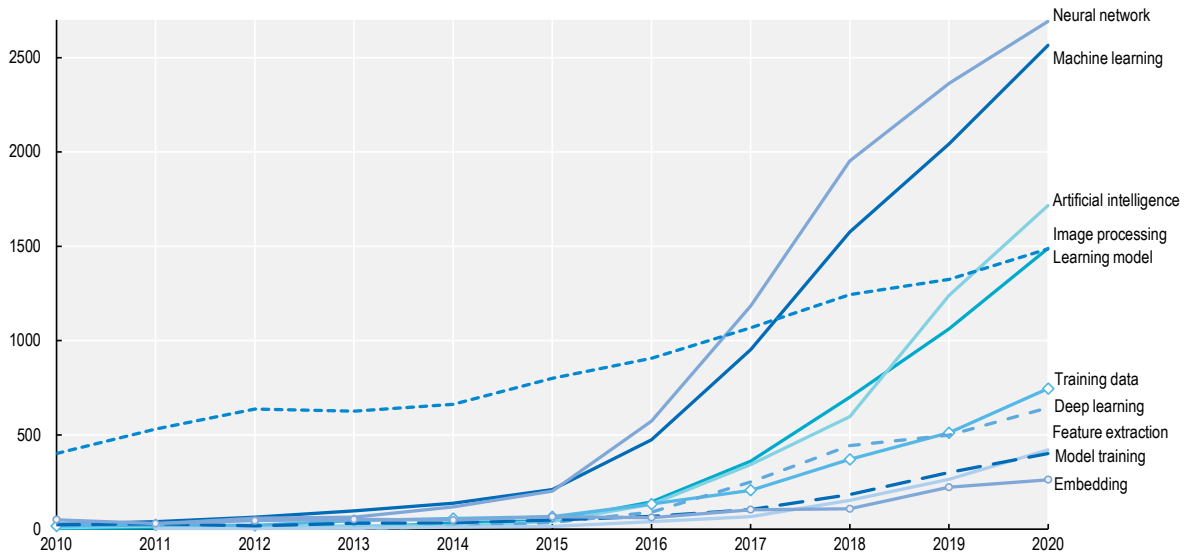
Note: Data refer to patent applications filed under the Patent Cooperation Treaty (PCT), by priority date. Figures for 2021-23 are estimated (see Annex E).
Source: OECD, STI Micro-data Lab: Intellectual Property Database, http://oe.cd/ipstats, January 2025.

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Estimates of the recent trends in AI patents, based on recently published patent records, show a marked slowdown since 2020, with an average growth of 2% a year in 2020-23. Recent data indicates that the global patenting also experienced a slight decrease since 2020, at an average yearly rate of -0.4% between 2020 and 2023. After filing of patents in the field of AI exploded since 2015, reaching a very high level in 2020, the stability observed during the last period nevertheless reflects sustained inventive activity in the field. The spectacular advances in AI patents seen through the 2010s have occurred in specific areas – notably in disruptive technologies such as generative AI and large language models (e.g., ChatGPT) – while the bulk of AI – applications to industrial processes – has continued to increase at a slower pace. In 2020, the top AI key phrases retrieved in patents included machine learning techniques, artificial intelligence in general and image processing (Figure 4.2).

Figure 4.2. Top 10 keywords listed in patents' titles or abstract, 2020

Number of patents by AI keyword



Note: Data refer to patent applications filed under the Patent Cooperation Treaty (PCT), by priority date and AI keywords, using whole counts.
Source: OECD, STI Micro-data Lab: Intellectual Property Database, http://oe.cd/ipstats, January 2025.

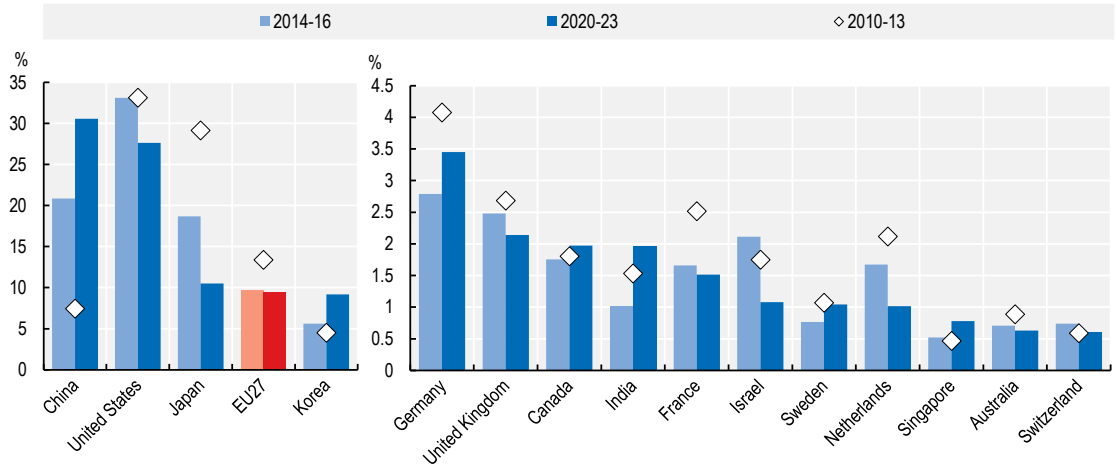
Leading countries in AI patents

The origin of inventions in AI is highly concentrated in a few countries: inventors from China, the United States and Japan contributed to 69% of AI patents in 2020-23 (Figure 4.3). China's share in total patents increased rapidly since the early 2010s, becoming the global leader with 30.6% of all AI patents worldwide. The United States, which had historically ranked first, saw its contribution decrease from 33% of all total AI patents in the 2010-13 to 27% in 2020-23. In turn, Japan's share dropped markedly from 29% in the early 2010s to 10.5% in the last period. The EU27 average dropped by four percentage points as compared to its level in 2010-13 but remained stable since 2014. Korea, Germany, Canada, India and Sweden are among the few countries that increased their shares of AI patents since the mid-2010s.

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Figure 4.3. Top 15 countries patenting in AI, 2020-23

Share of countries in total AI patents, PCT



Note: Data refer to patent applications filed under the Patent Cooperation Treaty (PCT), by priority date and inventor's location, using fractional counts. Figures for 2020-23 are estimated (see Annex E).
Source: OECD, STI Micro-data Lab: Intellectual Property Database, http://oe.cd/ipstats, January 2025.

The revealed technology advantage (RTA) indicators measure a country's relative specialisation in a given technology. The RTA index is calculated as the share of AI patents in a country's total AI patents, divided by the country's share of total global patents:

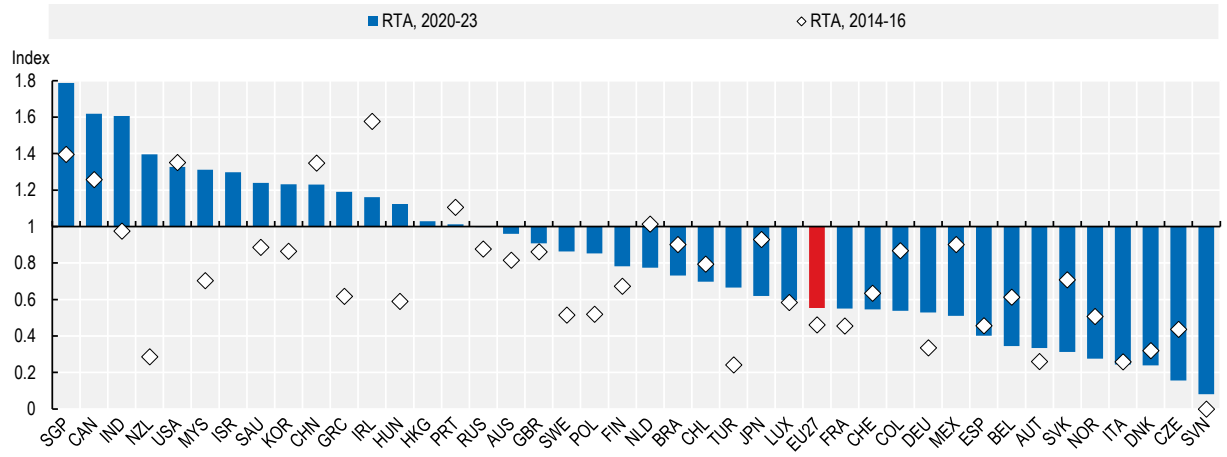
$$RTA_{ct} = \frac{\frac{P_{ct}}{\sum_c P_{ct}}}{\frac{P_c}{\sum_c P_c}}$$ **Equation 1**

where P_{ct} represents the number of patents filed by country *c* in technology *t,* and P_c refers to the total number of patents filed by country *c*. An RTA value above one indicates a relative advantage in AI, while a value below one suggests a relative disadvantage.

Singapore, Canada and India are the top three countries with the highest AI technological advantage based on the 2020-23 estimates and experienced an increase in their RTA since the mid-2010s (Figure 4.4). New Zealand displays a strong specialisation in AI patents in the last period, after being among the most under-specialised countries in 2014-16. The United States and China remain rather specialised in AI patenting. In contrast, Japan and EU27 economies show a relative under-specialisation in AI.

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Figure 4.4. Revealed technological advantage in AI, 2014-16 and 2020-23



Note: Data refer to patent applications filed under the Patent Cooperation Treaty (PCT), by priority date and inventor's location, using fractional counts. Only countries with more than 200 patents in the periods considered are included. Figures for 2020-23 are estimated (see Annex E).
Source: OECD, STI Micro-data Lab: Intellectual Property Database, http://oe.cd/ipstats, January 2025.

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5 Conclusion and limitations

The revised AI patent identification methodology presented in this paper expands coverage of AI inventions by incorporating the latest terminology, allowing for the inclusion of recent innovations such as generative AI. It uses a combined CPC and keyword approach, making it adaptable to various patent datasets. The method for identifying AI vocabulary and CPC classes is designed to enable easy and frequent updates (about twice a year). By operating at an aggregate CPC level, it remains stable despite potential classification changes, which typically occur at more granular levels. Such methodological approach can be applied to capture the development of emerging technology fields protected by patent documents in AI and beyond.

The new methodology for identifying AI patents faces several limitations. One significant challenge is the timeliness of patent information, as there are often administrative delays in patent publication. To mitigate this issue, the methodology includes a nowcasting approach to estimate AI patents for more recent years. Another limitation is the presence of blind spots regarding non-patented innovations, which are difficult to address due to the lack of available data. This highlights the challenges in capturing the full scope of AI inventions beyond what is formally patented. Patent-based measures may be complemented with indicators derived from other sources (e.g. scientific articles, open-source software, …) to provide a more comprehensive picture of the development of AI technologies.

The revised identification of AI patents presented in the paper will enable further exploration related to the impact of AI technologies, notably on knowledge spillovers spurring from AI technologies, on the relative value of those technologies for subsequent developments, on the ecosystem on which AI technologies relies, and on how AI overlaps with different technology domains.

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Annex A. Identify CPC codes related to AI

Table A A.1. Description of selected CPC subclasses connected to AI-related technologies

G06F	ELECTRIC DIGITAL DATA PROCESSING (computer systems based on specific computational models G06N)
G06K	GRAPHICAL DATA READING (image or video recognition or understanding G06V); PRESENTATION OF DATA; RECORD CARRIERS; HANDLING RECORD CARRIERS
G06N	COMPUTING ARRANGEMENTS BASED ON SPECIFIC COMPUTATIONAL MODELS
G06Q	INFORMATION AND COMMUNICATION TECHNOLOGY [ICT] SPECIALLY ADAPTED FOR ADMINISTRATIVE, COMMERCIAL, FINANCIAL, MANAGERIAL OR SUPERVISORY PURPOSES; SYSTEMS OR METHODS SPECIALLY ADAPTED FOR ADMINISTRATIVE, COMMERCIAL, FINANCIAL, MANAGERIAL OR SUPERVISORY PURPOSES, NOT OTHERWISE PROVIDED FOR
G06T	IMAGE DATA PROCESSING OR GENERATION, IN GENERAL
B64U	UNMANNED AERIAL VEHICLES [UAV]; EQUIPMENT THEREFOR
G16B	BIOINFORMATICS i.e., INFORMATION AND COMMUNICATION TECHNOLOGY [ICT] SPECIALLY ADAPTED FOR GENETIC OR PROTEIN-RELATED DATA PROCESSING IN COMPUTATIONAL MOLECULAR BIOLOGY
G16C	COMPUTATIONAL CHEMISTRY; CHEMOINFORMATICS; COMPUTATIONAL MATERIALS SCIENCE
G16H	HEALTHCARE INFORMATICS i.e., INFORMATION AND COMMUNICATION TECHNOLOGY [ICT] SPECIALLY ADAPTED FOR THE HANDLING OR PROCESSING OF MEDICAL OR HEALTHCARE DATA
G16Y	INFORMATION AND COMMUNICATION TECHNOLOGY SPECIALLY ADAPTED FOR THE INTERNET OF THINGS [IoT]
G16Z	INFORMATION AND COMMUNICATION TECHNOLOGY [ICT] SPECIALLY ADAPTED FOR SPECIFIC APPLICATION FIELDS, NOT OTHERWISE PROVIDED FOR
G06J	HYBRID COMPUTING ARRANGEMENTS (optical hybrid computing devices G06E3/00; {fuzzy computing G06N7/02}; neural networks for image data processing G06T; analog/digital conversion, in general H03M1/00)
G06V	IMAGE OR VIDEO RECOGNITION OR UNDERSTANDING
B60W	CONJOINT CONTROL OF VEHICLE SUB-UNITS OF DIFFERENT TYPE OR DIFFERENT FUNCTION; CONTROL SYSTEMS SPECIALLY ADAPTED FOR HYBRID VEHICLES; ROAD VEHICLE DRIVE CONTROL SYSTEMS FOR PURPOSES NOT RELATED TO THE CONTROL OF A PARTICULAR SUB-UNIT

Source: EPO and USPTO, Cooperative Patent Classification, https://www.cooperativepatentclassification.org/

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Table A A.2. Patent counts by CPC groups and AI-related keywords

CPC group	Total PCT patents	One AI keyword	2 AI keywords	More than 3 keywords	No AI keyword	% at least 1 AI keyword
G06N 3*	21682	6233	3491	5171	6787	69
G06N 20*	13615	3475	1511	2583	6046	56
G06F 18*	13507	3471	1576	2139	6321	53
G06N 7*	2920	698	331	510	1381	53
G06N 5*	5939	1438	606	1004	2891	51
G06V 10*	14338	3508	1305	1747	7778	46
G06V2201*	3146	775	293	358	1720	45
G06T 5*	4981	1520	336	361	2764	45
B64U 80	289	119	5	0	165	43
B64U 60	197	79	3	1	114	42
B64U 20	677	270	13	0	394	42
G06T 3*	3426	904	213	303	2006	41
B64U2201	1343	515	31	7	790	41
G06V 30*	2028	447	168	187	1226	40
G06T2201	109	34	3	6	66	39
B64U2101*	1887	667	44	15	1161	38
G06N 99	237	44	20	26	147	38
G06F 40*	6182	1128	521	591	3942	36
G06T2207*	18942	4218	1102	1498	12124	36
B64U 70	501	171	8	1	321	36
B64U 10*	2286	782	36	2	1466	36
G16H 30*	3645	662	231	399	2353	35
G06F2218	668	123	45	66	434	35
G06V 20*	13541	2812	849	1008	8872	34
G06T2211	321	56	28	26	211	34
G06F2207	217	43	11	20	143	34
G06T 7*	22857	4869	1228	1476	15284	33
B64U 30	1123	357	13	1	752	33
B64U 50	1231	375	17	2	837	32
G06T 1*	2099	469	97	87	1446	31
G16B 40*	1641	217	112	178	1134	31
G06J 1*	20	4	0	2	14	30
B60W 60*	2732	596	100	83	1953	29
G16B 15	357	41	18	41	257	28
G06T 9	1397	249	52	77	1019	27
G16C 60	194	22	10	20	142	27
G16C 20*	711	86	30	74	521	27
G06V 40*	10239	1706	459	559	7515	27
G06F 17*	3755	467	213	296	2779	26
G16H 50*	9157	1171	476	717	6793	26
G06T 11*	3784	622	156	197	2809	26
G16H 70	941	117	55	57	712	24
G06T2200	2303	376	84	95	1748	24
B60W2400*	50	9	2	1	38	24
G16B 35	180	13	9	21	137	24
B60W2420	1610	270	60	50	1230	24
G06F2216	119	18	3	7	91	24
G16B 10	84	9	3	7	65	23
B60W2554*	1761	293	64	34	1370	22
G16B 5	446	46	22	28	350	22

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CPC group	Total PCT patents	One AI keyword	2 AI keywords	More than 3 keywords	No AI keyword	% at least 1 AI keyword
G16H 15*	1123	129	46	63	885	21
G06F 5	197	19	9	13	156	21
G06T 17*	3197	424	107	119	2547	20
G06T2210	1938	253	57	76	1552	20
B60W2556*	1957	287	50	38	1582	19
B60W2555*	728	105	11	17	595	18
G06F 7	1454	146	50	66	1192	18
B60W2754	156	19	7	2	128	18
G06T 15	2620	330	49	90	2151	18
G16B 20	1479	120	54	90	1215	18
G16H 10*	4392	408	153	197	3634	17
G06F 16*	20840	2077	686	816	17261	17
B60W2050*	2455	312	55	41	2047	17
B60W 40*	3031	355	76	65	2535	16
G16C 10	104	12	0	5	87	16
G06T 13	1104	118	34	27	925	16
G16B 25	594	40	26	30	498	16
G06Q 99	44	3	2	2	37	16
G16B 45	171	12	5	10	144	16
G06T2219	1509	175	28	34	1272	16
B60W2756	149	18	1	4	126	15
G16B 50	549	37	19	28	465	15
B60W 50*	3563	403	64	70	3026	15
B60W2040	438	43	9	14	372	15
G06F 30	4989	402	151	177	4259	15
B60W 30*	4335	467	82	64	3722	14
G16B 30	1049	64	25	55	905	14
G16H 80	950	88	20	22	820	14
B60W2720*	761	82	12	9	658	14
B60W2552*	1285	130	21	18	1116	13
G06T 19*	4941	481	67	95	4298	13
B60W2520	1517	145	31	18	1323	13
B60W2540*	1570	145	30	24	1371	13
G16H 20*	6094	440	128	188	5338	12
G06Q 10*	17526	1267	387	440	15432	12
G06F2209	1259	82	20	48	1109	12
G06F2111	1157	79	30	28	1020	12
G06F 15*	1395	89	24	52	1230	12
B60W2422	68	8	0	0	60	12
G16H 40*	6720	462	129	192	5937	12
G16Z 99	88	4	1	5	78	11
G06T2215	207	16	3	4	184	11
G06Q 40	4693	264	102	143	4184	11
G06F2119	1007	58	22	27	900	11
G06Q 50*	13817	911	256	269	12381	10
G06Q 30*	12884	813	216	297	11558	10
G16Y 20	344	22	8	5	309	10
B64U 40	40	4	0	0	36	10
G16Y 40*	588	30	17	11	530	10
G06F2113	802	48	11	20	723	10

Note: PCT patents in 2018-21, by filing date and CPC groups, using whole counts. CPC groups marked with * were validated by experts.
Source: OECD, STI Micro-data Lab: Intellectual Property Database, http://oe.cd/ipstats, January 2025.

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Annex B. New AI search strategy

Table A B.1. Revised list of AI-related keywords

action recognition	activity recognition	adaboost	adaptive boosting	adversarial learning
adversarial network	ai algorithm	ai chipset	algorithmic of ai	ambient intelligence
answer set program	ant colony	ant colony optimisation	artificial bee colony algorithm	artificial evolution
artificial intelligence	artificial neural network	association rule	attention (…) transformer	attention based
autoencoder	automated planning	automated scheduling	autonomic computing	autonomous car
autonomous vehicle	autonomous weapon	backpropagation	bayes network	Bayesian learning
bayesian network	bee colony	blind signal separation	bootstrap aggregation	brain computer interface
brownboost	classification tree	cluster analysis	cognitive automation	cognitive computing
cognitive insight system	cognitive modelling	cognitive robotic	collaborative filtering	collision avoidance
community detection	computational intelligence	computational pathology	computer vision	connectionis
convolutional neural	convolutional neural network	decision model	decision tree	deep belief network
deep convolutional neural network	deep learning	deep learning model	deep neural	deep neural network
deep reinforcement learning	dictionary learning	differential evolution algorithm	diffusion model	dimensionality reduction
driverless car	driverless vehicle	dynamic time warping	embedding	emotion recognition
ensemble learning	evolutionary algorithm	evolutionary computation	evolutionary robotic	expert system
extreme machine learning	face recognition	facial expression recognition	factorisation machine	feature engineering
feature extraction	feature learning	feature selection	federated learning	firefly algorithm
fuzzy c	fuzzy environment	fuzzy logic	fuzzy number	fuzzy set
fuzzy system	gaussian mixture model	gaussian process	generation model	generative adversarial network
generative ai	genetic algorithm	genetic programming	gesture recognition	gradient boosting
gradient tree boosting	graph convolutional network	graph neural	graph neural network	graphical model
gravitational search algorithm	hebbian learning	hidden markov model	hierarchical clustering	high dimensional data
high dimensional feature	high dimensional input	high dimensional model	Hough transformer	human action recognition
human activity recognition	human aware artificial intelligence	human robot interaction	humanoid robot	hybrid ai
image classification	image processing	image recognition	image retrieval	image segmentation
independent component analysis	inductive logic	inductive logic programm	inductive monitoring	instance based learning
intelligence augmentation	intelligent agent	intelligent classifier	intelligent geometric computing	intelligent infrastructure
intelligent software agent	intelligent system	intuitionistic fuzzy set	K means	Kernel learning
language (…) transformer	language model	large language	large language model	latent dirichlet allocation
latent semantic analysis	latent variable	layered control system	learning (…) transformer	learning algorithm
learning automata	learning model	link prediction	llm	logistic regression
logitboost	long short term memory	lpboost	machine intelligence	machine learning
machine learning model	machine translation	machine vision	madaboost	MapReduce
Markovian	memetic algorithm	meta learning	model training	motion planning

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multi agent system	multi label classification	multi layer perceptron	multi objective evolutionary algorithm	multi objective optimisation
multi sensor fusion	multi task learning	multilayer perceptron	multinomial naïve Bayes	naïve Bayes classifier
natural gradient	natural language	natural language generation	natural language processing	natural language understanding
nearest neighbour algorithm	neural architecture search	neural machine translation	neural network	neural network cnn
neural network model	neural radiance field	neural turing	neural turing machine	neuromorphic architecture
neuromorphic computing	neuromorphic engineering	non negative matrix factorisation	object detection	object recognition
obstacle avoidance	one class learning	one shot learning	optical character recognition	optimisation for learning
optimization for learning	particle swarm optimisation	pattern recognition	pedestrian detection	planning algorithm
policy gradient method	Q learning	random field	random forest	rankboost
recommender system	recurrent neural network	regression tree	reinforcement learning	relational learning
representation learning	robot interaction	robot learning	rough set	rule based learning
rule learning	self driving car	self organising map	self organising structure	semi supervised learning
semi supervised training	sensor data fusion	sensor fusion	sentiment analysis	service robot
similarity learning	simultaneous localisation mapping	single linkage clustering	sparse representation	spectral clustering
speech recognition	speech synthesis	speech to text	stacked generalisation	statistical relational learning
stochastic gradient	stochastic gradient descent	supervised learning	supervised training	support vector machine
support vector regression	swarm intelligence	swarm optimization	t s fuzzy system	Takagi Sugeno fuzzy system
temporal difference learning	text mining	text to speech	topic model	totalboost
training data	trajectory planning	trajectory tracking	transfer learning	trust region policy optimisation
unmanned aerial vehicle	unsupervised learning	unsupervised training	variational inference	vconvolutional neural
vector machine	virtual assistant	visual servoing	xgboost	

Note: The list of keywords was elaborated using the most recent keyword list extracted from arXiv in May 2024.
Source: OECD, 2025

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Table A B.2. Revised list of AI-related CPC groups

Figures in parenthesis indicate the share of patents allocated to a given CPC group featuring at least one AI related keyword in the title or abstract

All patents with CPC	G06N3 (69%)	G06N20 (56%)	G06F18 (53%)	G06N7 (53%)	G06N5 (51%)
Patents with CPC and at least one AI-related keyword in the title of abstract	B60W30 (14%)	B60W40 (16%)	B60W50 (15%)	B60W60 (29%)	B60W2040 (15%)
	B60W2050 (17%)	B60W2400 (24%)	B60W2420 (24%)	B60W2422 (12%)	B60W2520 (13%)
	B60W2540 (13%)	B60W2552 (13%)	B60W2554 (22%)	B60W2555 (18%)	B60W2556 (19%)
	B60W2720 (14%)	B60W2754 (18%)	B60W2756 (15%)	B64U10 (36%)	B64U20 (42%)
	B64U30 (33%)	B64U40 (10%)	B64U50 (32%)	B64U60 (42%)	B64U70 (36%)
	B64U80 (43%)	B64U2101 (38%)	B64U2201 (41%)	G06F5 (21%)	G06F7 (18%)
	G06F15 (12%)	G06F16 (17%)	G06F17 (26%)	G06F30 (15%)	G06F40 (36%)
	G06F2111 (12%)	G06F2113 (10%)	G06F2119 (11%)	G06F2207 (34%)	G06F2209 (12%)
	G06F2216 (24%)	G06F2218 (35%)	G06J1 (30%)	G06N99 (38%)	G06Q10 (12%)
	G06Q30 (10%)	G06Q40 (11%)	G06Q50 (10%)	G06Q99 (16%)	G06T1 (31%)
	G06T3 (41%)	G06T5 (45%)	G06T7 (33%)	G06T9 (27%)	G06T11 (26%)
	G06T13 (16%)	G06T15 (18%)	G06T17 (20%)	G06T19 (13%)	G06T2200 (24%)
	G06T2201 (39%)	G06T2207 (36%)	G06T2210 (20%)	G06T2211 (34%)	G06T2215 (11%)
	G06T2219 (16%)	G06V10 (46%)	G06V20 (34%)	G06V30 (40%)	G06V40 (27%)
	G06V2201 (45%)	G16B5 (22%)	G16B10 (23%)	G16B15 (28%)	G16B20 (18%)
	G16B25 (16%)	G16B30 (14%)	G16B35 (24%)	G16B40 (31%)	G16B45 (16%)
	G16B50 (15%)	G16C10 (16%)	G16C20 (27%)	G16C60 (27%)	G16H10 (17%)
	G16H15 (21%)	G16H20 (12%)	G16H30 (35%)	G16H40 (12%)	G16H50 (26%)
	G16H70 (24%)	G16H80 (14%)	G16Y20 (10%)	G16Y40 (10%)	G16Z99 (11%)

Note: Percentages are based on the set of patent applications filed under the Patent Cooperation Treaty (PCT) in 2018-21.
Source: OECD, STI Micro-data Lab: Intellectual Property Database, http://oe.cd/ipstats, January 2025.

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Annex C. Expert consultation

Elements of the revised AI patent identification methodology were submitted to a group of experts in AI, composed of 11 AI experts and seven patent examiners, for validation. This was done using online surveys composed of the preliminary lists of AI-related vocabulary, preselected CPC groups and a random sample set of patents. 13 submissions were received.

Based on this feedback, the following decisions were made:

- The proposed revised list of keywords was kept;
- CPC categories were revised by going one level deeper in the classification tree to better capture specific details at a granular level.

Keywords

Experts were asked to categorise keywords as "relevant to AI", "context dependent", "not related to AI" and "can't tell". Among the 290 keywords reviewed, a large majority were classified as either "relevant to AI" or "context dependent".

No keyword received more than three "not related to AI". Five keywords received three: "*industrial robot*", "*legged robot*", "*point cloud*" (a robot related technology), "*robot*", "*robot fleet*". Another nine keywords received two "not related to AI": "*biped robots*", "*cyber physic systems*", "*high dimensional space*", "*high dimensional system*", "*quadruped robot*", "*semantic web*", "*optical character extraction*", "*social robot*", and "*wheeled mobile robot*". Almost all keywords that were rated lower by some experts related to robotics or referred to broader concepts (e.g., "*high dimensional system*"). These keywords were removed from the list.

Since the keywords will be contextualised by the CPC categories when compiling statistics, any overinclusive biases are expected to be limited. Therefore, it is recommended to keep them in the final list.

CPC categories

Experts were asked to classify 13 CPC subclasses as "core AI", "related but not core", "unrelated" or "can't tell". The subclass G06N was unanimously classified as "core AI", while G06T and G06V, both related to image processing and understanding, received three "core AI" votes. The remaining subclasses were mostly classified as "related but not core", with one expert noting that G06F18/00 is "core AI", while other G06F subcategories are "related but not core".

AI patent validation

Experts examined the titles and abstracts of 220 patents (divided into four modules of 55 patents each) from the selected CPC groups. Responses revealed a lack of agreement among experts regarding the boundaries of AI patents. For example, in module A the two experts agreed on only 14 of the 55 proposed patents; in module B, the four experts agreed on 13 out of 55 patents, and three experts agreed on another 18.

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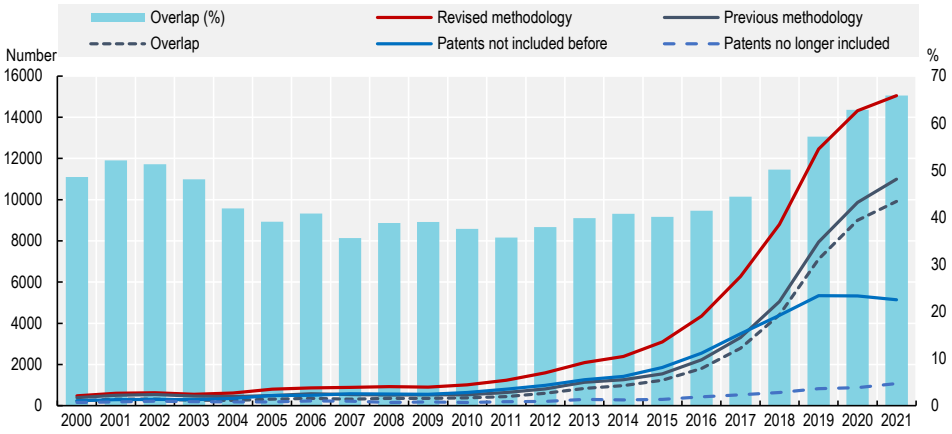
Experts' responses were used to assess the list of proposed categories. Overall, 80% of experts agreed that patents from the core CPC groups were AI-related. This percentage amounted to 58% for the AI groups combining CPC and keywords and 33% for patents drawn from close groups (same subclass but different group). The correlation between experts' ratings and the proportion of patents with at least one AI related keyword was 0.44, which is statistically significant. Hence, based on experts' judgement, the proposed AI patent classification is considered statistically valid.

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Annex D. Comparison with previous methodology

A comparison was conducted between patent statistics derived from the 2020 OECD taxonomy (Baruffaldi et al., 2020[1]) and indicators compiled on the revised AI patent identification methodology. The number of AI patents increased by over 50% under the revised methodology, with nearly 60% of patents identified by both methodologies (Figure A D.1).

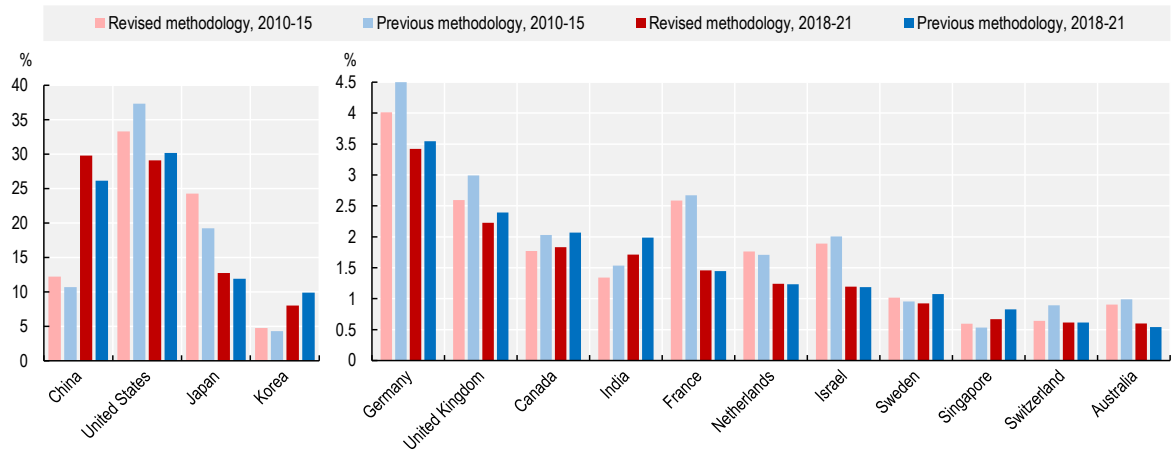
Figure A D.1. Previous vs. revised OECD AI patents identification methodology



Note: Data refer to patent applications filed under the Patent Cooperation Treaty (PCT), by filing date and inventor's location, using fractional counts. The previous methodology for identifying AI patents relies on Baruffaldi et al. (2020[1]).
Source: OECD, STI Micro-data Lab: Intellectual Property Database, http://oe.cd/ipstats, January 2025.

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Figure A D.2. Country's share in total AI patents, previous vs. revised OECD methodology, 2010-15 and 2018-21

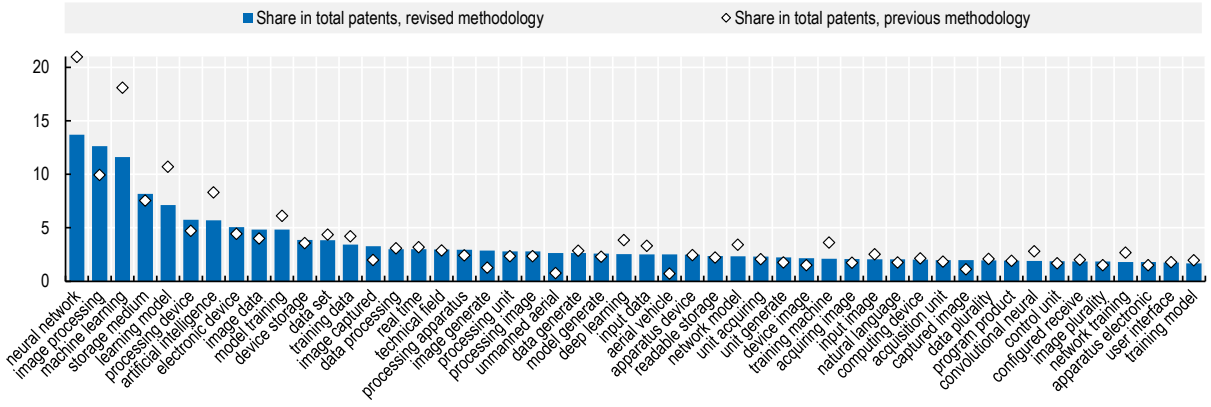


Note: Data refer to patent applications filed under the PCT, by filing date and inventor's location, using fractional counts. The previous methodology for identifying AI patents relies on Baruffaldi et al. (2020[1]).
Source: OECD, STI Micro-data Lab: Intellectual Property Database, http://oe.cd/ipstats, January 2025.

The differences mainly stem from the inclusion of additional AI keywords, in particular those related to generative AI techniques and image processing, and the removal of restrictions on the minimum number of keywords required in patent's title or abstract (in the 2020 OECD methodology, combinations of at least two AI keywords were imposed). Country shares in AI patent development remain consistent across methodologies (Figure A D.2), with over 98% correlation between the two methods.

Figure A D.3. Top 30 frequent terms emerging from AI patent abstracts, previous vs. revised methodology

Bigram



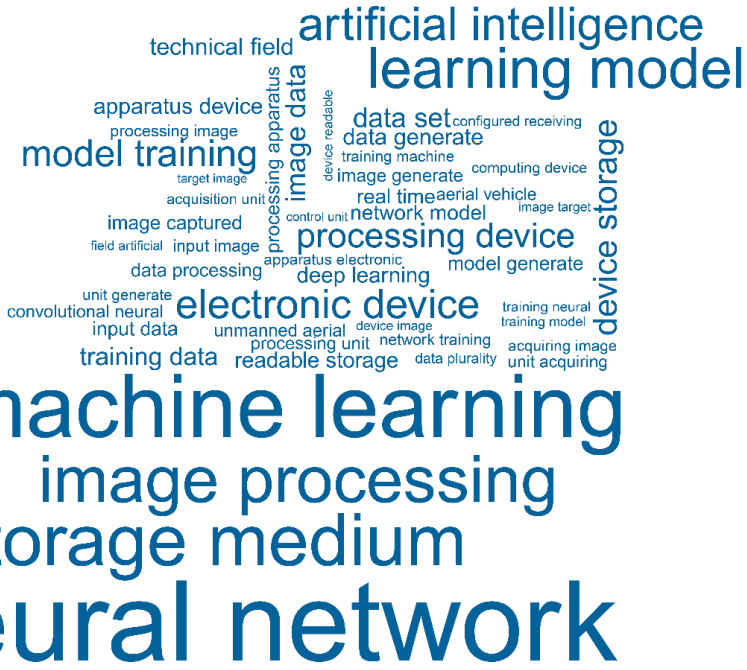
Note: Data refer to patent applications filed under the Patent Cooperation Treaty (PCT). The previous AI patent identification methodology relies on Baruffaldi et al. (2020[1]). Top frequent bigram terms have been extracted from the patents abstract, using text mining techniques as described in Box 3.1.
Source: OECD, STI Micro-data Lab: Intellectual Property Database, http://oe.cd/ipstats, January 2025.

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A complementary text mining analysis of patent abstracts reveal that core AI techniques, such as neural networks and machine learning, remain prevalent (Figure A D.3). However, the revised revised methodology captures a broader range of inventions related to data storage, image processing, and data processing (Figure A D.4). Additionally, there is a noticeable increase in patents related to unmanned aerial vehicles.

Figure A D.4. Word cloud of AI patents' abstract, 2018-21

Top terms extracted from AI patents' abstracts, bigrams



Note: Data refer to patent applications filed under the Patent Cooperation Treaty (PCT). Top frequent bigram terms have been extracted from the patents abstract, using text mining techniques as described in Box 3.1.
Source: OECD, STI Micro-data Lab: Intellectual Property Database, http://oe.cd/ipstats, January 2025.

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Annex E. Method for nowcasting AI PCT patents

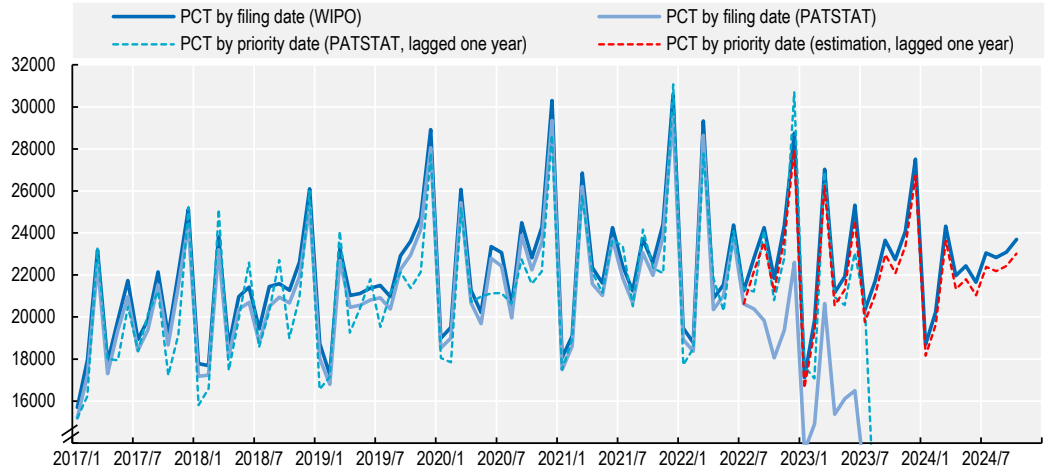
Statistics on patent counts traditionally face timeliness issues. Indicators are calculated according to the earliest date at which the invention has been protected by a patent – i.e., the priority date - which is the closest date to when the invention took place. There is a minimum 18-month legal delay before the patent can be publicly disclosed, to which a delay of one additional year may be added owing to the Paris Convention (OECD, 2009[14]). Hence, in 2024, data available in the Spring 2024 edition of the EPO PATSTAT database are incomplete for patent applications filed under the Patent Cooperation Treaty (PCT) from priority year 2021. Indicators on total PCT patents and AI patents are "nowcasted" up to 2023, using partial data available from alternative sources of data.

Estimation of total PCT patents

The World Intellectual Property Organization (WIPO) publishes monthly statistics on the number of PCT patent filings received, based on unpublished records, on its IP Statistics Data Centre.[12] In January 2025, monthly patent counts published by the WIPO were provided up to October 2024. As shown in Figure A E.1, trends in WIPO's monthly statistics are aligned with patent counts compiled using the PATSTAT database (Spring 2024 edition).

Figure A E.1. Statistics on PCT filings by different sources

Counts of patents by filing date or by priority date (lagged one year), WIPO vs. PATSTAT



Note: Data refer to counts of PCT patents by filing month or priority date. Counts by priority dates are displayed with a lag of one year: data point for 2020/1 refer to priority month 2019/1.
Sources: OECD, STI Micro-data Lab: Intellectual Property Database, http://oe.cd/ipstats, and WIPO, IP Statistics Data Centre, January 2025.

Noteworthy, a differential of about 3% on average per month is observed between WIPO and PATSTAT counts until filing month July 2022. Differences are due to patent filings that never get public (withdrawn by the applicant before publication). Furthermore, monthly trends by filing date are highly correlated with

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those observed by priority date, with a one-year delay: the correlation coefficient between counts by filing date and counts by priority date - lagged by one year - amount to 0.95 for the period 2017/1 to 2021/7.

Therefore, the latest trends in PCT patents by filing date can be used to extrapolate the total number of patents by priority date, lagging by one year, from July 2021 to October 2023. Finally, as the proportion of patents filed up to the month of October in total patents per year is stable over time, it is applied to estimate the total number of patents for priority year 2023.

Estimation of AI patents

WIPO also publishes patent documents on its PATENTSCOPE database,[13] available on-line. PATENTSCOPE provides access to published international patent applications in full-text, on the day of the publication. It offers access to latest published patent records not yet included in the EPO PATSTAT database. The advanced search module of the platform permits the retrieval of PCT patents for a given priority date, and to search for key phrases in the text of the patent. It was used to count the number of PCT patents published by priority date with at least one AI keyword mentioned in the abstract.

Statistics reported in Table A E.1 were pulled from PATENTSCOPE for the period 2015-23, for the full years and the first 6 months of each year, to accommodate for truncation in the last year. This was done for total PCT patents, and for a set of AI patents identified with a selection of AI keywords. The search was performed using a selection of top 20 frequent AI keywords identified in AI patents, as observed in the EPO PATSTAT during the period 2018-23.[14]

Table A E.1. PCT patent counts for a set AI keywords, based on PATENTSCOPE

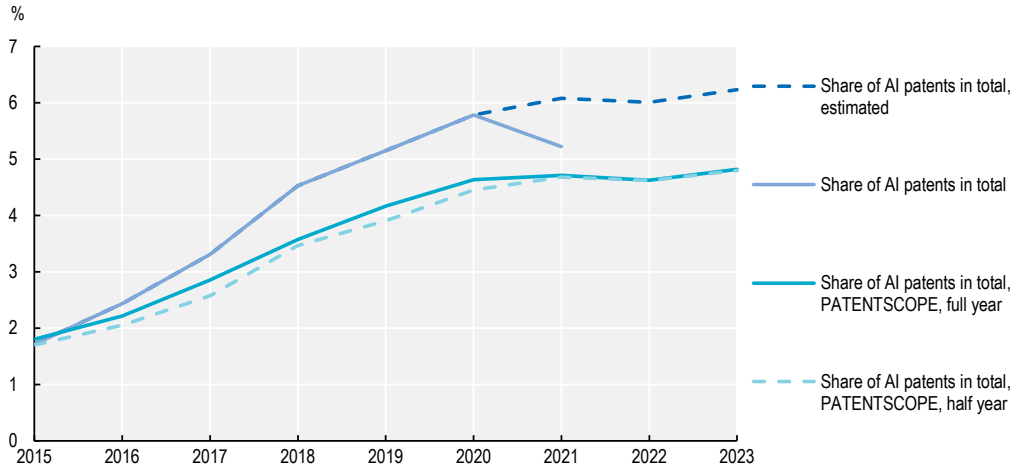
	PCT patents, AI keyword, full year	PCT patents, AI keyword, up to June of the year	PCT patents, total full year	PCT patents, total up to June of the year
2015	3 735	1 726	207 098	101 453
2016	4 808	2 221	216 542	107 973
2017	6 527	2 922	228 494	113 524
2018	8 555	4 038	239 295	116 474
2019	10 173	4 702	244 170	120 395
2020	11 901	5 486	256 786	123 260
2021	11 985	5 882	254 236	125 662
2022	11 413	5 585	246 605	120 803
2023	6 550	5 751	135 811	119 869

Source: WIPO, PATENTSCOPE database, accessed on 27 January 2025.

Counts derived from PATENTSCOPE can be used to extrapolate the evolution of the share of AI patents in total. The share of AI patents identified with a set of keywords follows a similar trend to that of the observed number of AI patents in total PCT using PATSTAT (Figure A E.2). The growth rate of AI share in PCT patents based on PATENTSCOPE can therefore be applied to extrapolate the share of AI patents in total. The share of AI patents is then applied on the estimated total PCT patent counts to nowcast AI patents up to 2023.

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Figure A E.2. Evolution of the share of AI patents in total, by sources



Note: The share of AI patents in total is based on data available in PATSTAT. The share of AI patents extracted from PATENTSCOPE is based on patents selected using a set of 20 AI keywords.
Sources: OECD, STI Micro-data Lab: Intellectual Property Database, http://oe.cd/ipstats, January 2025, and WIPO, PATENTSCOPE database, accessed on 27 January 2025.

Robustness tests were performed on the period 2018-20 to assess the validity of the methodology: about 2 percentage points difference are observed between the “true” figures compiled for those year and the ones estimated with the nowcasting technique.

In a final step, nowcasted AI patents are distributed among inventor countries, based on the average country shares observed during the period 2020-22. Information on inventors appear to be partially lacking in PATENTSCOPE for the most recent years, as patent applicants do not always share detailed information on inventors’ team when the patent is first published. Hence the latest country shares are extrapolated using PATSTAT data compiled over the period 2020-22: the country distribution for 2021 corresponds to the distribution observed in 2021 given the high number of observations for this year. 2022 shares are based on the average shares observed during the period 2020-22 and 2023 shares are based on the average observed for 2021 and 2022 estimates.

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Endnotes

[1] https://www.questel.com/patent/ip-intelligence-software/orbit-intelligence/

[2] https://www.epo.org/en/searching-for-patents/business/patstat

[3] https://patentsview.org/

[4] https://eto.tech/dataset-docs/country-ai-activity-metrics/

[5] The classifier was trained on labeled seed patents – GenAI - and negative seeds - non-GenAI AI patents

[6] https://arxiv.org/

[7] https://www.kaggle.com/datasets/Cornell-University/arxiv

[8] For further information on the CPC, see: https://www.cooperativepatentclassification.org/obj.

[9] In the CPC structure, subclasses refer to categories of “level 3”, like G06N, while groups refer to categories of “level 4”, like G06N5. Hence, each subclass comprises several groups.

[10] The term “Transformer” is on the one hand central to AI - notably generative AI – and, on the other hand, extremely polysemic, as it is also associated to electrical technologies that are unrelated to AI. Hence it was necessary to keep the term in key phrases but ensure its contextual disambiguation, which was done thanks to CPC categories and to complementary terms (i. e. “attention”, “language”, “learning”) retrieved in the surrounding words (see Table A B.1).

[11] The patent classification being performed by patent office examiners (human experts in the domain), being classified in these five groups must be reflective of the dominance of AI technology.

[12] https://www3.wipo.int/ipstats/key-search/indicator

[13] https://www.wipo.int/en/web/patentscope

[14] The top frequent AI keywords present in PCT AI patents during the period 2018-23 are : "neural network", "machine learning", "image processing", "learning model", "artificial intelligence", "training data", "deep learning", "convolutional neural", "unmanned aerial vehicle", "model training", "feature extraction", "autonomous vehicle", "learning algorithm", "natural language", "object detection", "embedding", "artificial neural network", "generation model", "image recognition", "deep neural".

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